A Few Notes on Vector Calculus

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1 Introduction

TBD

2 Notation

Some of the notation we'll use is described in this section.

2.1 Vectors

Some authors use bold font to denote a vector. Adding a "hat" denotes a unit vector. A unit vector is defined to be

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

that is, $\hat{\mathbf{u}}$ is a vector of unit length in the $\mathbf{u}$ direction.

For example $\hat{\mathbf{i}}$ denotes the unit vector in the x direction. Using these definitions the parametric form of a three dimensional vector $\mathbf{r}(t)$ would be

$$\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$$

In this case $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$, the basis vectors in $\mathbb{R}^3$, have column vector format

$$\hat{\mathbf{i}} = \begin{bmatrix}1\\0\\0\end{bmatrix},\ \hat{\mathbf{j}} = \begin{bmatrix}0\\1\\0\end{bmatrix},\ \hat{\mathbf{k}} = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

On the other hand, some authors use "over arrow"

$$\vec{r}(t) = g(t)\,\vec{i} + h(t)\,\vec{j} + k(t)\,\vec{k}$$

I have also seen

$$\vec{r}(t) = g(t)\,\hat{i} + h(t)\,\hat{j} + k(t)\,\hat{k}$$

In this document I'll use these forms interchangeably.

Definition 2.1. Dot Product: The dot product (or scalar product or inner product) of two row vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, where $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, is defined to be the scalar value

$$\mathbf{a} \cdot \mathbf{b} = \langle \mathbf{a}, \mathbf{b} \rangle = \sum_{j=1}^{n} a_j b_j = \mathbf{a}\mathbf{b}^{\intercal}$$

According to the Pythagorean theorem, the length of a vector $\mathbf{a} = (a_1, a_2, a_3) \in \mathbb{R}^3$, denoted $\|\mathbf{a}\|$, equals $\sqrt{a_1^2 + a_2^2 + a_3^2}$. The next definition is a generalization of the notion of length to vectors of R^n.

Definition 2.2. Euclidean Norm: The Euclidean norm of a vector

$$\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$$

is defined to be

$$\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\sum_{j=1}^{n} a_j^2} \tag{1}$$

Theorem 2.1. Cauchy–Schwarz inequality:

$$\left| \langle \mathbf{a}, \mathbf{b} \rangle \right| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

Proof: The inequality is trivial if either $\mathbf{a}$ or $\mathbf{b}$ is zero, so assume that neither is. Now, let $\mathbf{x} = \frac{\mathbf{a}}{\|\mathbf{a}\|}$ and $\mathbf{y} = \frac{\mathbf{b}}{\|\mathbf{b}\|}$ and so $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$ ($\mathbf{x}$ and $\mathbf{y}$ are unit vectors). Then we can see by the triangle inequality [4, 7] that

$$
\begin{aligned}
\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| &\leq \|\mathbf{x}-\mathbf{y}\| && \text{\# triangle inequality [4, 7]} \\
&\Rightarrow 0 \leq \|\mathbf{x}-\mathbf{y}\| && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1,\ 1-1=0,\ |0|=0 \\
&\Rightarrow 0 \leq \|\mathbf{x}-\mathbf{y}\|^2 && \text{\# } \|\mathbf{x}-\mathbf{y}\| \geq 0 \Rightarrow \|\mathbf{x}-\mathbf{y}\|^2 \geq 0 \\
&\Rightarrow 0 \leq \langle \mathbf{x}-\mathbf{y}, \mathbf{x}-\mathbf{y} \rangle && \text{\# } \|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a} \rangle \text{ by Equation (1)} \\
&\Rightarrow 0 \leq \sum_{j=1}^{n} (x_j - y_j)^2 && \text{\# by Equation (1)} \\
&\Rightarrow 0 \leq \sum_{j=1}^{n} \left(x_j^2 - 2x_j y_j + y_j^2 \right) && \text{\# expand previous line} \\
&\Rightarrow 0 \leq \sum_{j=1}^{n} x_j^2 - 2\sum_{j=1}^{n} x_j y_j + \sum_{j=1}^{n} y_j^2 && \text{\# sum is a linear operator} \\
&\Rightarrow 0 \leq \|\mathbf{x}\|^2 - 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2 && \text{\# by Equation (1)} \\
&\Rightarrow 0 \leq 1 - 2\langle \mathbf{x}, \mathbf{y} \rangle + 1 && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1 \Rightarrow \|\mathbf{x}\|^2 = \|\mathbf{y}\|^2 = 1 \\
&\Rightarrow 0 \leq 2 - 2\langle \mathbf{x}, \mathbf{y} \rangle && \text{\# } 1+1=2 \\
&\Rightarrow -2 \leq -2\langle \mathbf{x}, \mathbf{y} \rangle && \text{\# subtract 2 from both sides} \\
&\Rightarrow 1 \geq \langle \mathbf{x}, \mathbf{y} \rangle && \text{\# divide by -2, flipping inequality} \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y} \rangle \leq 1 && \text{\# rearrange} \\
&\Rightarrow \left\langle \frac{\mathbf{a}}{\|\mathbf{a}\|}, \frac{\mathbf{b}}{\|\mathbf{b}\|} \right\rangle \leq 1 && \text{\# } \mathbf{x} = \frac{\mathbf{a}}{\|\mathbf{a}\|},\ \mathbf{y} = \frac{\mathbf{b}}{\|\mathbf{b}\|} \\
&\Rightarrow \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\|\|\mathbf{b}\|} \leq 1 && \text{\# scalar multiplication rule for dot products: } \langle c_1\mathbf{a}, c_2\mathbf{b} \rangle = c_1 c_2 \langle \mathbf{a}, \mathbf{b} \rangle \\
&\Rightarrow \langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\|\|\mathbf{b}\| && \text{\# multiply both sides by } \|\mathbf{a}\|\|\mathbf{b}\|
\end{aligned}
$$

Next note that the absolute value of x, $|x|$, is defined to be

$$
|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0. \end{cases}
$$

If we replace $\mathbf{a}$[1] by $-\mathbf{a}$ and since by Equation (1) we have $\|-\mathbf{a}\| = \|\mathbf{a}\|$ and by the scalar multiplication rule for dot products with $c_1 = -1$ and $c_2 = 1$ we have $\langle -\mathbf{a}, \mathbf{b} \rangle = -\langle \mathbf{a}, \mathbf{b} \rangle$ we can see that

$$
-\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|
$$

We also know from above that

$$
\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|
$$

So by the definition of absolute value we have

$$
\big|\langle \mathbf{a}, \mathbf{b} \rangle\big| \leq \|\mathbf{a}\|\|\mathbf{b}\|
$$

□

[1] There is nothing special about using $\mathbf{a}$ here. We could use $\mathbf{b}$ and get the same result.

2.2 Derivatives

We will also denote the derivative of a vector $\mathbf{r}$ with respect to t by $\dot{\mathbf{r}}$, $\dfrac{d\mathbf{r}}{dt}$ or $\mathbf{r}'(t)$. Note also that by definition

$$\frac{d\mathbf{r}}{dt} = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

2.3 Vector Fields

In vector calculus and physics, a vector field is an assignment of a vector to each point in a subset of space [2]. For example, a vector field in the plane can be visualized as a collection of arrows with a given magnitude and direction, each attached to a point in the plane. Vector fields are often used to model the speed and direction of a moving fluid throughout space, or the strength and direction of some force, such as the magnetic or gravitational force, as it changes from one point to another point.

For example, the wind vector field for the 2011 Joplin, MO tornado [5] is shown in Figure 1. Here the color represents the wind speed $\|\mathbf{v}\|$, where $\mathbf{v}$ is the wind velocity vector.

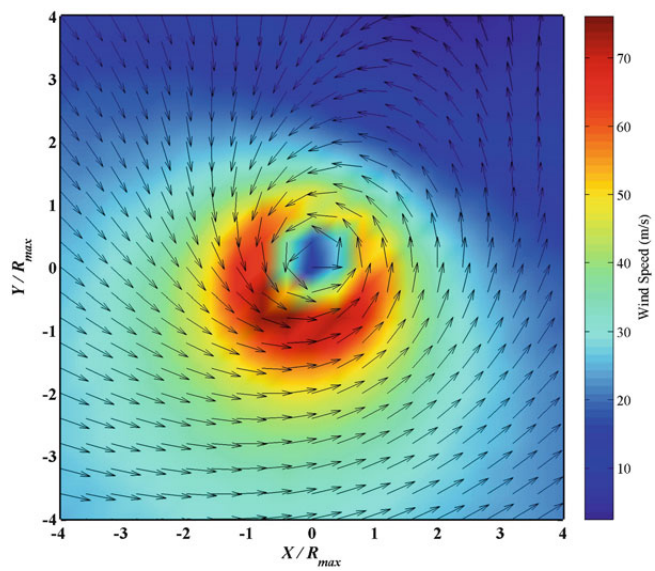

Figure 1: 2011 Joplin, MO Tornado Wind Velocity Vector Field

3 Arc Length

Consider a segment of a parametric curve $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$ between two points $P = \mathbf{r}(t)$ and $Q = \mathbf{r}(t + \Delta t)$, as shown in Figure 2.

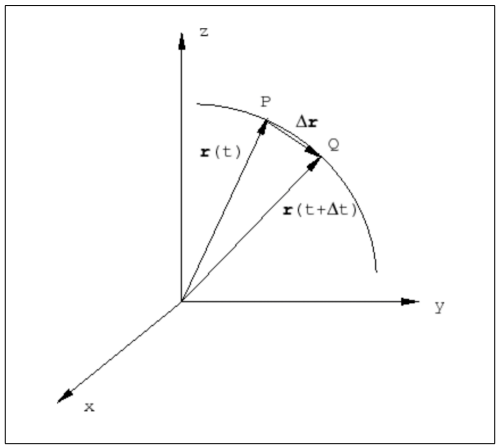

Figure 2: $\Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$

What is the length of the segment of the curve between P and Q? This length is called the arc length and is denoted by Δs. Δs can be approximated by the chord length $\|\Delta\mathbf{r}\|$. Then we can compute the infinitesimal arc length ds as follows:

$$
\begin{aligned}
\Delta s &\approx \|\Delta\mathbf{r}\| && \text{\# approximate } \Delta s \text{ by } \|\Delta\mathbf{r}\| \\
&= \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\| && \text{\# } \Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t) \\
&= \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\| \cdot \frac{\Delta t}{\Delta t} && \text{\# multiply by } 1 = \frac{\Delta t}{\Delta t} \\
&= \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# assume } \Delta t > 0 \\
\Rightarrow\ & \Delta s \approx \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# this is our approximation} \\
\Rightarrow\ & \lim_{\Delta t \to 0} \Delta s = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# take the limit of both sides} \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# } \Delta t \to 0 \Rightarrow \Delta s \to 0 \Rightarrow \lim_{\Delta t \to 0} \Delta s = ds \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# product rule for limits [1]} \\
\Rightarrow\ & ds = \left\| \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# for a normed vector space } X \ \lim \|x_n\| = \|\lim x_n\| \\
\Rightarrow\ & ds = \left\| \frac{d\mathbf{r}}{dt} \right\| \cdot dt && \text{\# } \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt},\ \lim_{\Delta t \to 0} \Delta t = dt
\end{aligned}
$$

Interestingly we can also see this using the Pythagorean theorem. Specifically, if we move an infinitesimal distance in the x and y directions we get a triangle in the x-y plane who's sides have length dx and dy and who's hypotenuse is ds. This scenario is shown for some curve $f(x)$ in Figure 3.

In the two dimensional case where the parameterization is $x = g(t)$ and $y = h(t)$ we have $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}}$. Further, since derivative is a linear operator [6] we know that

$$\frac{d\mathbf{r}}{dt} = \frac{dg}{dt}\,\hat{\mathbf{i}} + \frac{dh}{dt}\,\hat{\mathbf{j}} \tag{2}$$

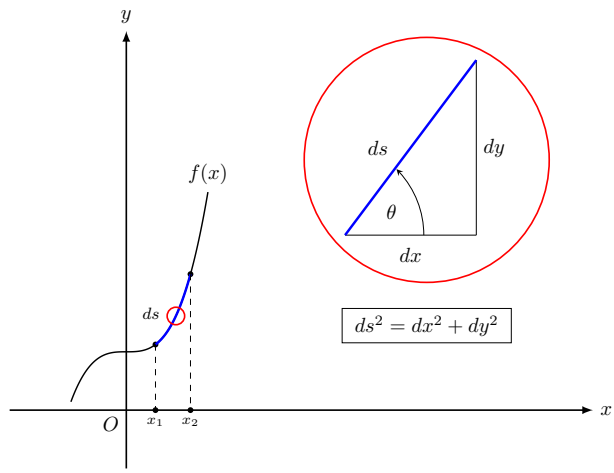

Figure 3: $f(x)$, ds and the Pythagorean Theorem

So in the source domain the Pythagorean theorem tells us that $ds = \sqrt{dx^2 + dy^2}$. Using this expression for ds we can see that

$$\begin{aligned}
ds &= \sqrt{dx^2 + dy^2} && \text{\# Figure 3} \\
&= \sqrt{dg^2 + dh^2} && \text{\# switch to parametric form: } x = g(t) \text{ and } y = h(t) \\
&= \sqrt{dg^2 + dh^2} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt},\ dt > 0 \\
&= \sqrt{dg^2 + dh^2} \cdot \left[\sqrt{\frac{1}{dt^2}} \cdot dt\right] && \text{\# } \frac{x}{x} = \sqrt{\left(\frac{1}{x}\right)^2} \cdot x = \sqrt{\frac{1}{x^2}} \cdot x,\ x > 0 \\
&= \sqrt{\left(dg^2 + dh^2\right) \cdot \left(\frac{1}{dt^2}\right)} \cdot dt && \text{\# simplify: } \sqrt{x} \cdot \sqrt{\frac{1}{y}} = \sqrt{x \cdot \frac{1}{y}} \\
&= \sqrt{\left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2} \cdot dt && \text{\# simplify: } \left(x^2 + y^2\right) \cdot \left(\frac{1}{z^2}\right) = \frac{x^2}{z^2} + \frac{y^2}{z^2} = \left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 \\
&= \sqrt{g'(t)^2 + h'(t)^2} \cdot dt && \text{\# switch notation from Leibniz} \rightarrow \text{Lagrange: } \frac{df}{dt} = f'(t) \\
&= \left\|\frac{d\mathbf{r}}{dt}\right\| \cdot dt && \text{\# Equation (2) and the definition of } \|\cdot\|
\end{aligned}$$

In any event, now we see that

$$\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\| \tag{3}$$

and so

$$\frac{ds}{dt} = \|\dot{\mathbf{r}}\| \tag{4}$$

Since $\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\| = \|\dot{\mathbf{r}}\|$ we also know that the unit tangent vector $\mathbf{T} = \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} = \frac{d\mathbf{r}}{ds}$ [3]. One way to see this is to notice that

$$\begin{aligned}
\mathbf{T} &= \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} && \text{\# definition of the unit tangent vector } \mathbf{T} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\|\dot{\mathbf{r}}\|} && \text{\# definition of } \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} && \text{\# Equation (4): } \|\dot{\mathbf{r}}\| = \frac{ds}{dt} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt} \\
&= \frac{d\mathbf{r}}{ds} && \text{\# simplify}
\end{aligned}$$

4 Conclusions

Acknowledgements

LaTeX Source

References

[1] Alex Svirin. Properties of Limits. `https://math24.net/properties-limits.html`. [Online; accessed 14-November-2021].

[2] Galbis, A. and Maestre, M. *Vector Analysis Versus Vector Calculus*. Universitext. Springer New York, 2012.

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